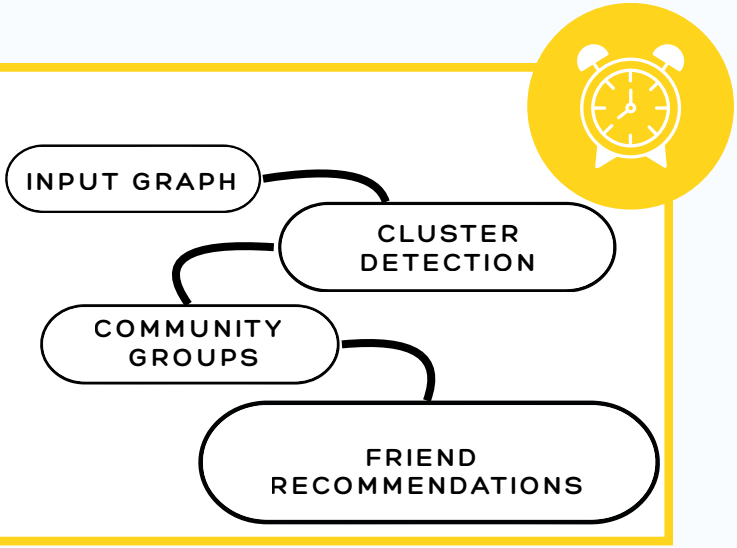
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Friend Suggestions using Graph Clustering

INTRODUCTION

Social networks rely on algorithms to suggest new friends to users based on shared connections and interests.

Using graph clustering, we can identify tightly connected groups of users (communities) and recommend friends from the same cluster.



ALGORITHM COMPLEXITY AND TYPE

Step	Operation	Time complexity
Build Graph	O(V+E)	o(v+E)
Cluster Detection	o(n log n)	O(n log n)
Suggestion	O(V)	O(V-i)

OUTPUT VISUALIZATON



Node A is in Cluster 1; suggests Node B and Node C (same cluster, not directly connected).

REAL-WORLD APPLICATIONS

- Social Media Platforms – Friend or Group Suggestions
- E-commerce – Product Recommendations
- Research Networks – Collaboration Suggestions
- Fraud Detection – Identify suspicious communities

CONCLUSION / REFERENCES

Graph clustering efficiently uncovers hidden connections in social networks, helping systems recommend relevant friends.

Future improvements could include weighted edges (based on message frequency or interests).

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